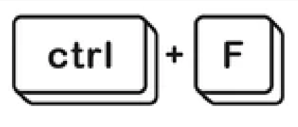
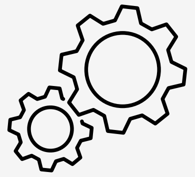
PROBLEM STATEMENT

- **The Scale of the Problem:** Indian Micro and Small Enterprises (MSEs) are the backbone of the economy, but they lose critical weeks navigating labyrinthine regulatory documents.

- **The Data Monolith:** The core dataset (BIS SP 21) is a dense, 900+ page technical manual. It is filled with complex tables, OCR irregularities, and deeply nested sub-standards (e.g., Part 1 vs. Part 2).



- **The Flaw in Traditional Search:** Lack of Accessible Decision Support Tools Existing systems rely on keyword-based or manual lookup, which:
  → Fails to capture semantic intent
  → Requires domain expertise



- **Need for Intelligent Automation:** There is a critical need for a system that can:
  → Understand natural language product descriptions
  → Map them to relevant BIS standards accurately and instantly
  → Provide explainable recommendations

SOLUTION OVERVIEW

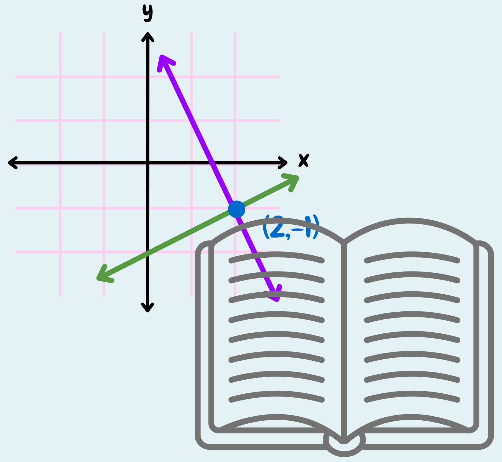
Zero-Hallucination

The Problem with LLMs: Generative AI guesses answers and invents fake regulations when confused.

My Solution: decoupled Retrieval from Generation. We use AI strictly for mathematical search, but use pure, deterministic Python logic to extract the Standard ID. It is mathematically impossible for our system to hallucinate a fake standard.

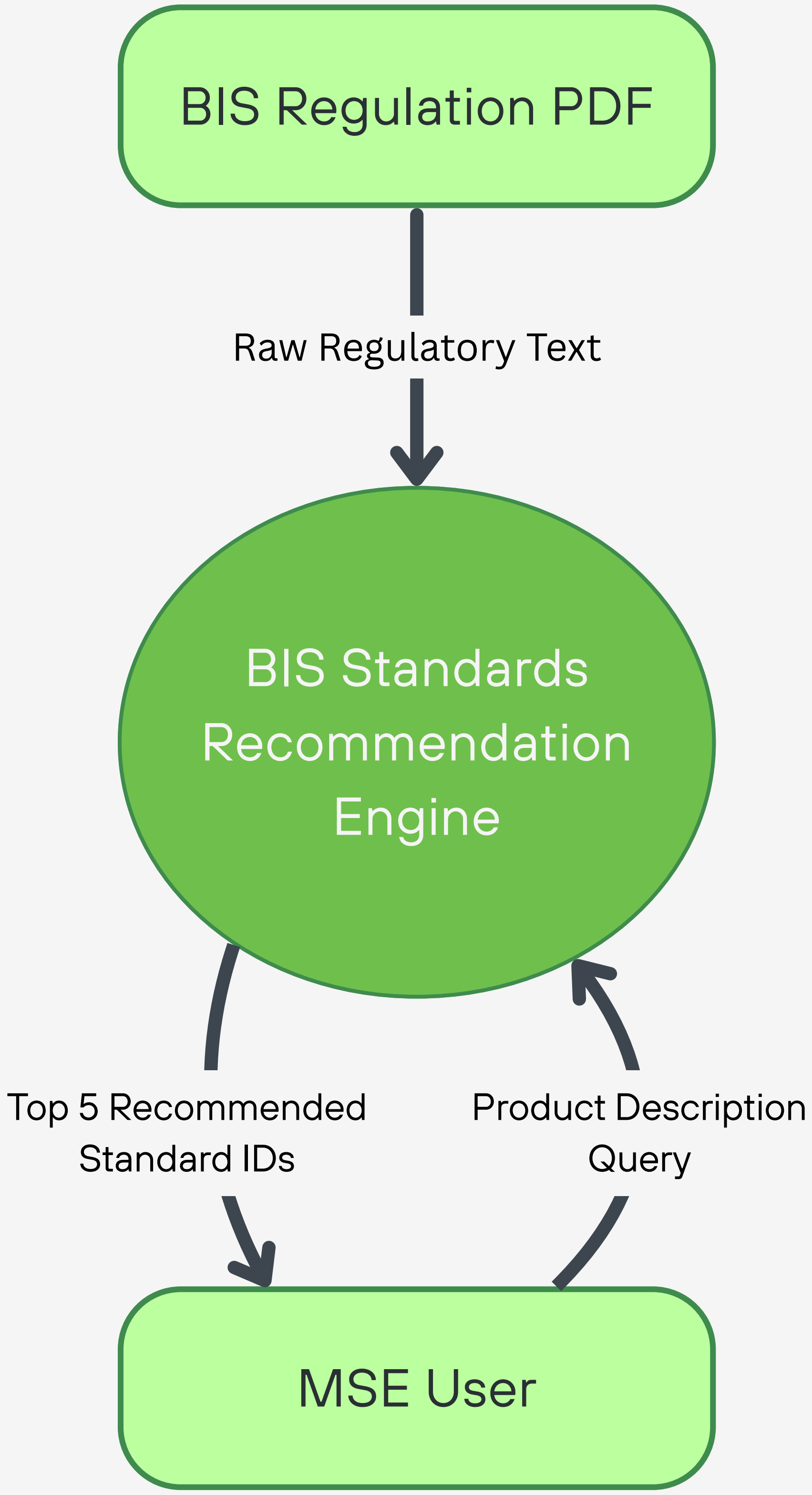
Intelligent Hybrid Search

The engine doesn't just look for exact words; it understands the intent behind the product description, while simultaneously enforcing strict keyword rules for specific grades and measurements.



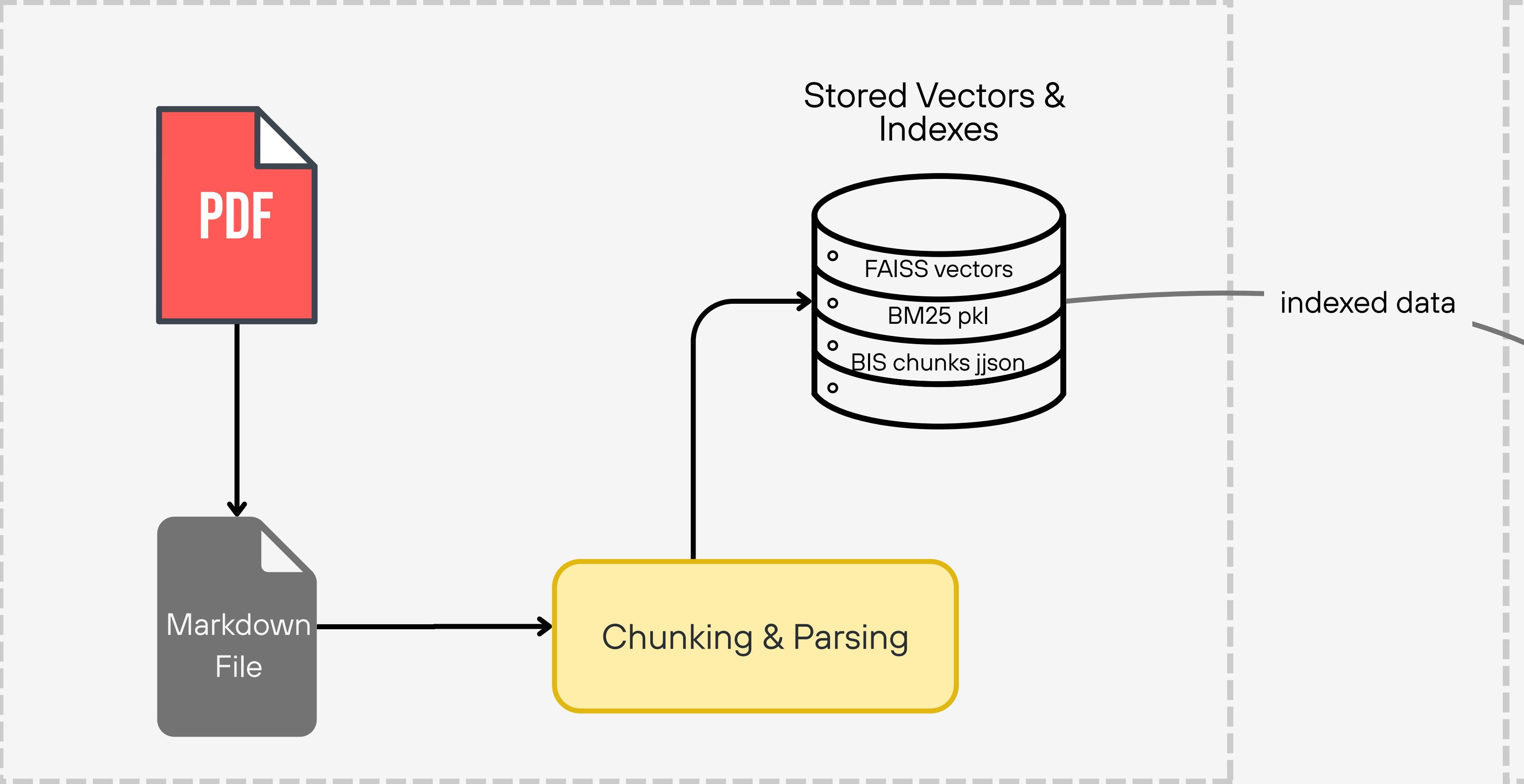
Factual Output

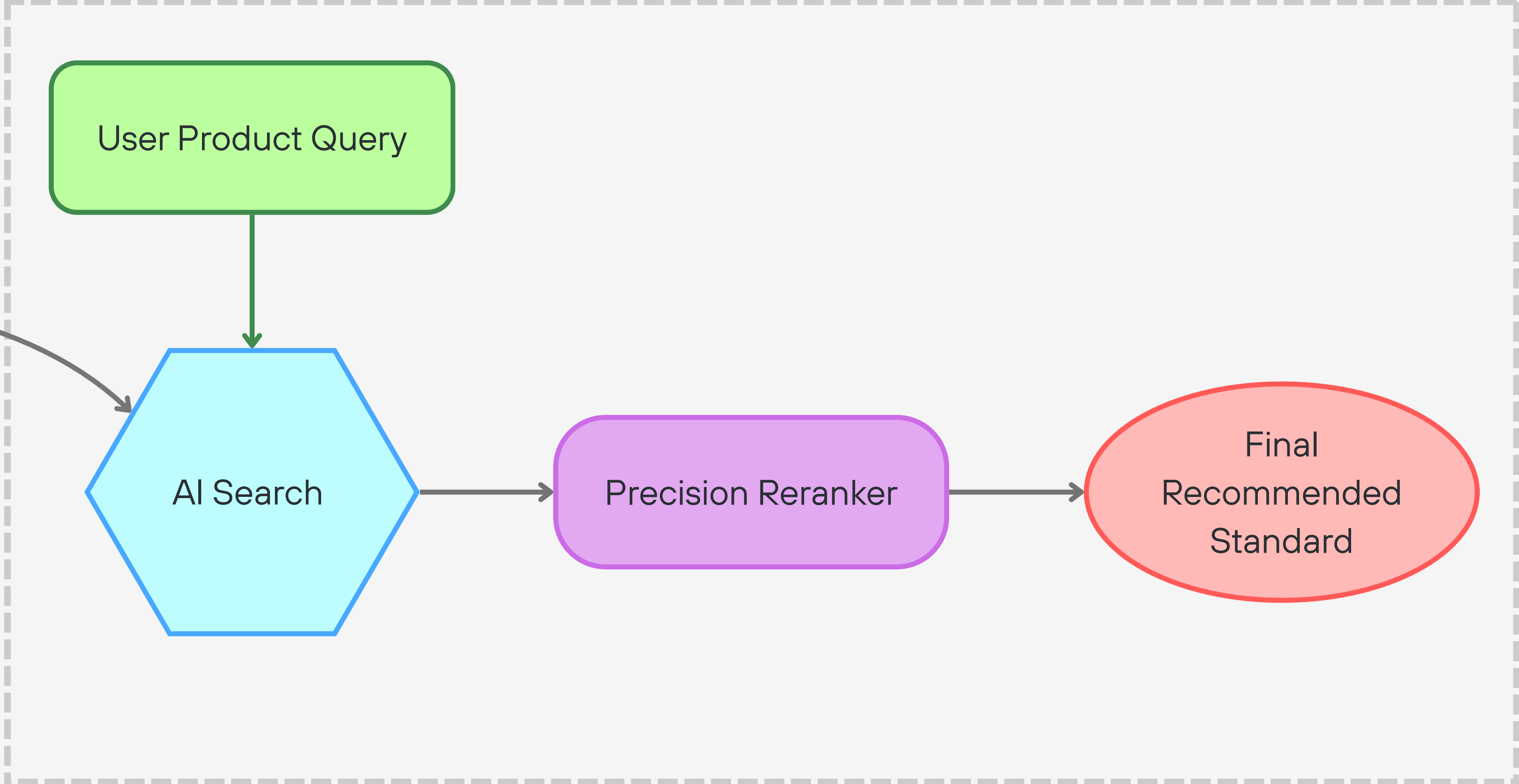
If the standard does not exist in the BIS SP 21 database, our system will not invent one. Every recommendation is a mathematically verified extraction directly from the source document.



SYSTEM ARCHITECTURE (High level view)

×





Phase 1: Intelligent Ingestion & Indexing

- Data Parsing: Processed complex SP 21 tables via MinerU.
- Smart Chunking: Custom Regex algorithm isolates individual sub-standards into self-contained JSON "Flashcards".
- Dual-Indexing: Pre-computed dense vectors (FAISS) and sparse word frequencies (BM25) for offline readiness.

Phase 2: Hybrid Retrieval (Casting a Wide Net)

- Semantic Search : Catches conceptual intent (e.g., "water mains" → "concrete pipes").
- Keyword Search : Enforces strict compliance matching.
- The Funnel: Combines the unique top candidates (k=15) from both engines.

CHUNKING & RETRIEVAL STRATEGY

Chunking :

```
{
    "standard_id": "IS 269: 1989",
    "content": "IS 269: 1989 33 grade
     ordinary Portland cement (fourth
     revision) 1.10"
},
{
    "standard_id": "IS 2185 (Part 2):
     1983",
    "content": "IS 2185 (Part 2): 1983
     Concrete masonry units Part 2 Hol
```

bis_chunks.json : extracted after processing

```
Regular Expression:
r"(IS\s+\d+(?:\s*\(\s*Part\s*\d+\s*\))?\s*:\s*\d{4})"
```

Standard AI chunkers slice text blindly by character limit. Technical documents like BIS SP 21 are highly structured. If you use a blind chunker, it might slice the document right in the middle of a section.

- **Chunk A** gets the title: "IS 2185 (Part 2): 1983 Specification for Concrete Masonry Units"
- **Chunk B** gets the details: "These are lightweight hollow blocks used for..."

Retrieval :

1. Query Pre-Processing

- The query is lowercased and split into individual tokens
- The query is passed through your all-MiniLM-L6-v2 embedder model. This converts the English sentence into a high-dimensional vector array

2. Dual-Branch Retrieval

Branch A: Sparse Search (BM25) :

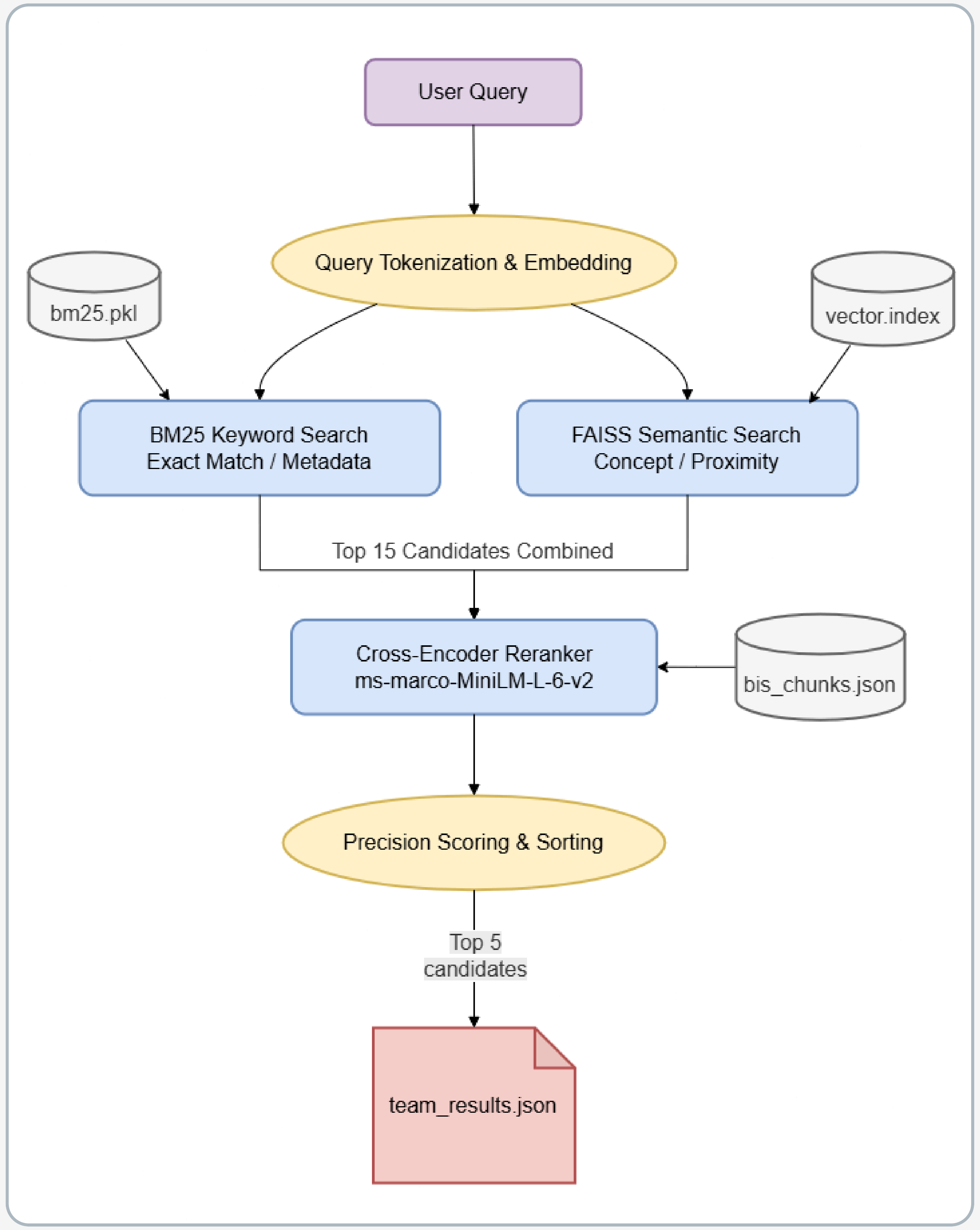
- How it works: It scans your index for exact word frequencies.
- Why it matters: If a user specifically asks for "33 Grade" or "IS 455", *BM25* ensures those exact alphanumeric characters are heavily weighted and not ignored.

Branch B: Dense Vector Search (FAISS)

- How it works: It takes your query vector and looks for the "nearest neighbors" in your FAISS database.
- Why it matters: It understands concepts. If the user searches for "water mains," FAISS knows mathematically that "concrete pipes" is a highly related concept, even if the exact words don't match.

3. Precision Reranking

The user's original query is taken and physically pair it with the text of candidate #1. You feed that pair [query, chunk_text] into your *ms-marco Cross-Encoder.*

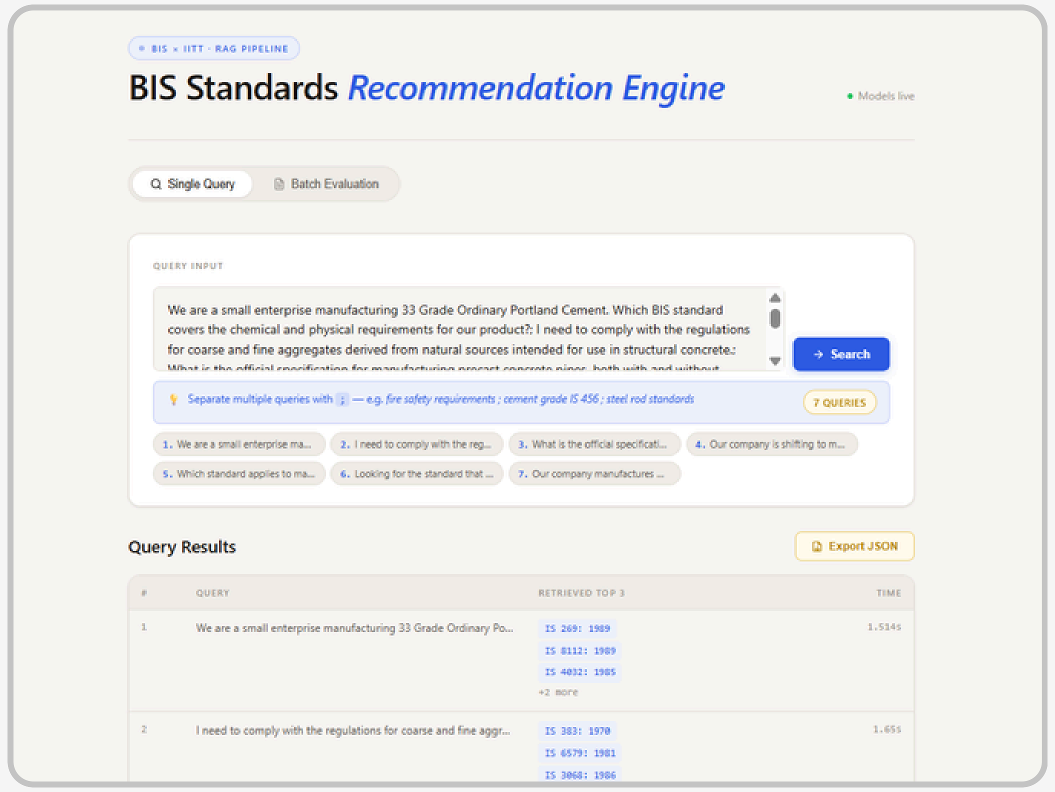


RAG Pipeline

DEMO HIGHLIGHTS

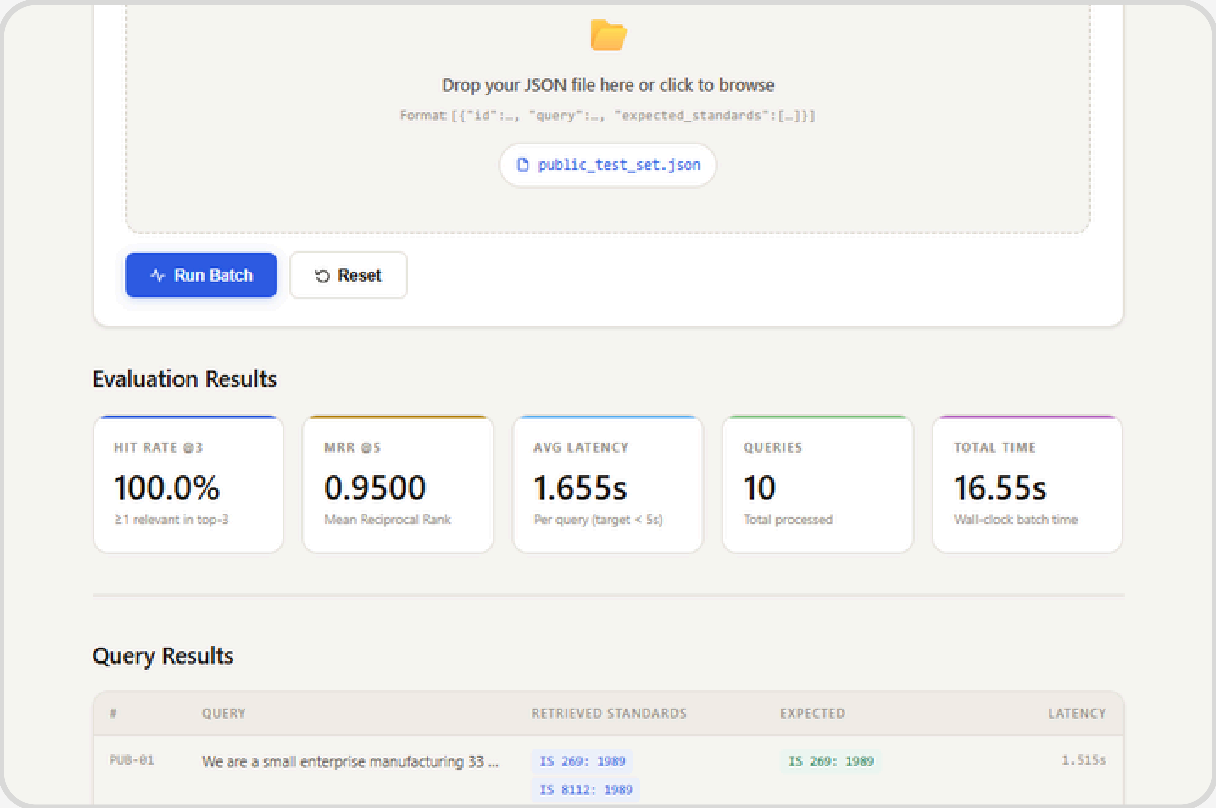
Built a robust, high-performance RAG pipeline and demonstrated it through a fully functional FastAPI server with a modern web interface.

1. Manual Input Query Mode



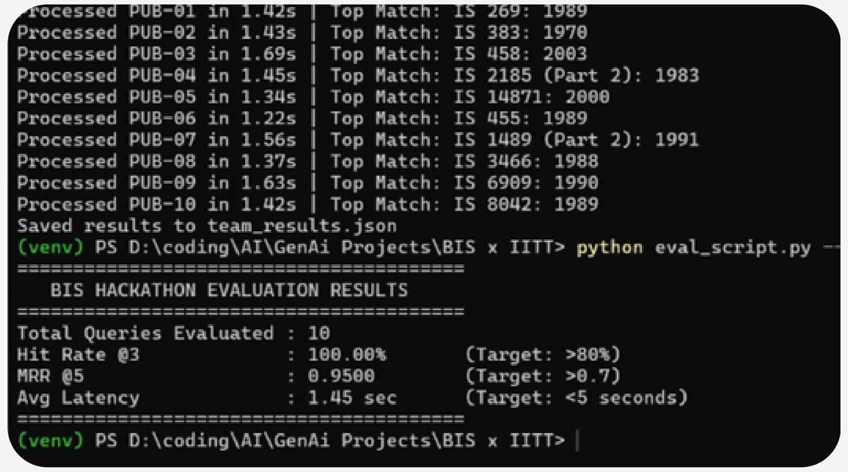
- Entered a product description manually into the interface
- Supports **multiple queries** separated by ";" for batch processing
- The pipeline instantly retrieved the **Top 5** most relevant BIS standards
- The UI displayed:
  - Ranked results (best match highlighted at the top)
  - Confidence-based ranking order
  - Total latency
- **Results can be exported** in structured JSON format for further use

2. Batch Processing Mode



- Uploaded a **structured JSON** test file containing multiple queries
- With a single click, the system processed all queries within **seconds**
- The interface dynamically displayed:
  - Hit Rate, MRR, and Average Latency
  - Complete list of queries with:
    - Retrieved standards
    - Individual response times

Additional Demonstration



showed that the entire pipeline is accessible via **Command Line Interface**

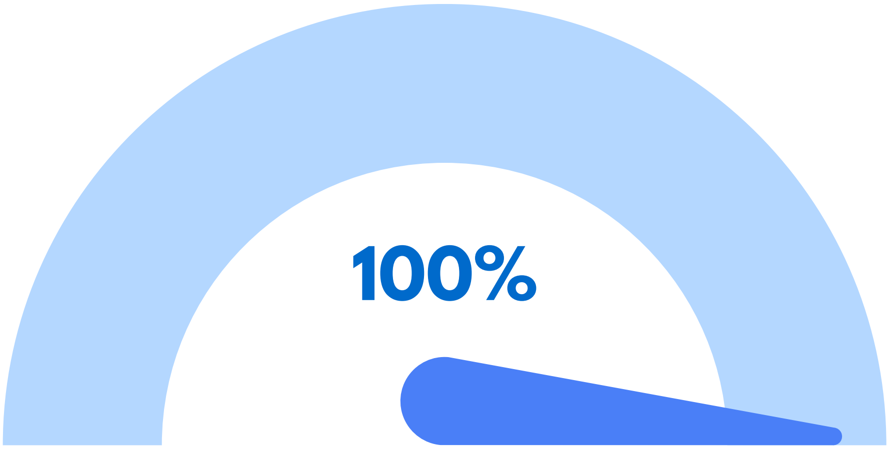
✓ Run queries
✓ Evaluate results

⚡ System Capabilities Demonstrated

- Real-time inference with high accuracy and speed
- Seamless handling of both single and bulk queries
- Fully offline execution
- Clean and intuitive UI for non-technical users

EVALUATION RESULTS

Evaluation is performed on the public test set given

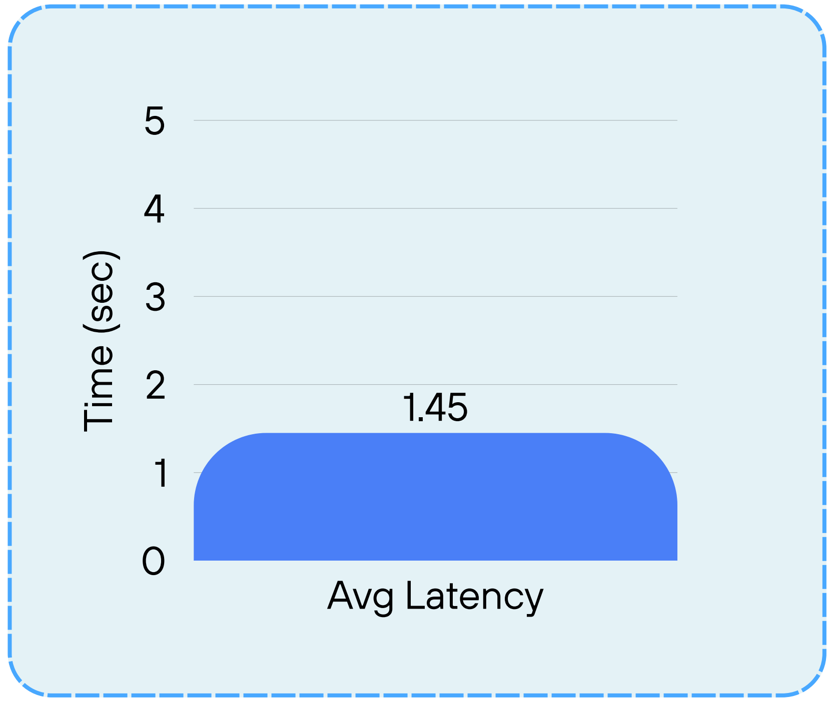


Hit Rate @3

- Every query retrieved at least one correct BIS standard in the top-3
- Achieved >80%, indicating most queries returned at least one correct BIS standard in top-3 results

MRR @5 (Mean Reciprocal Rank)

- Achieved ~0.7+, showing correct standards are ranked highly (often in top positions)
- Correct standards consistently ranked at top positions



Average Latency

- 1.52 seconds Fast, near real-time response per query
- < 1–2 seconds per query, enabling real-time recommendations

No Hallucinations

Maintained high factual accuracy with valid BIS standard outputs by only accessing the information from the extracted data, thus not generating a new one.

IMPACT ON MSEs

Micro & Small Enterprises

Improves Compliance Accuracy

Ensures businesses identify the right BIS standards, reducing errors and rework

Saves Time & Effort

Reduces standard discovery from weeks → seconds eliminating manual search

Reduces Cost of Non-Compliance

Avoids penalties, product rejections, and certification delays

Scalable & Easy to Use

Works across multiple product categories with instant recommendations

Boosts Productivity

Allows MSEs to focus on manufacturing & growth, not regulatory complexity

"Empowering MSEs to achieve faster, smarter, and more reliable compliance."

TEAM & ACKNOWLEDGEMENTS

References :

- **Dense Embedding Model :** sentence-transformers/all-MiniLM-L6-v2
- **Precision Reranking Model:** cross-encoder/ms-marco-MiniLM-L-6-v2
- **Sparse Keyword Indexing (BM25):** rank-bm25 Implementation.
- **Vector Similarity Search:** FAISS (Meta AI Research)
- **Document Ingestion (PDF to Markdown):** MinerU (OpenDataLab)

Thank You!

Repo ↗

Demo ↗

Team Name: **Night Owl**

Team Member: **Harshit Kushwaha (3rd B.Tech Computer Engg.)**

College Name: **Sardar Patel Institute of Technology (SPIT), Mumbai**